



Hospital Operations Workflow Optimization & EMR Architecture

End-to-End Business Analysis & Digital Transformation Blueprint

Prepared by: Rohit Shorey
Healthcare Systems Engineering



Operational Inefficiencies & Target Objectives

Current Bottlenecks:

- **90+ Min Wait Times:** Intake-to-consultation delays congest waiting areas.
- **Vulnerable Record Storage:** Manual paper files cause redundant data entry and file loss risks.
- **Inter-Departmental Silos:** Paper prescription slips lead to transit delays and fulfillment errors between doctors, pharmacy, and billing

Strategic Objectives & KPIs:

- **-40% Intake Wait Time:** Streamline patient intake through digital self-service check-in.
- **Unified Cloud EMR:** Enable instantaneous patient history lookup for attending clinical staff.
- **Direct e-Prescribing:** Eliminate transcript errors via real-time automated pharmacy queue routing.



Stakeholder Analysis Matrix

Stakeholder Group	Project Role	Influence	Interest	Key Requirement
Hospital Board	Sponsor / Budget	High	High	ROI, regulatory compliance, patient throughput.
Attending Doctors	Clinical End-Users	High	High	Sub-second medical history lookup, intuitive UI.
Nursing Staff	Operational Intake	Medium	High	Rapid vital signs entry, no paper clipboards.
Pharmacy / Lab	Fulfillment Leads	Medium	High	Legible e-prescriptions, automated inventory sync.
IT Operations	Infrastructure & Security	High	Medium	99.9% uptime SLA, HIPAA compliance, role access control.
Patients	Service Beneficiaries	Low	High	Reduced wait times, transparent billing, data privacy.

Current State Workflow & Operational Bottlenecks

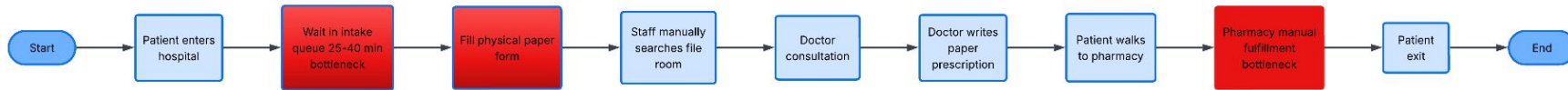
Hospital Patient Journey

Hospital Patient Journey

Owner: Rohit Shorey

Last Updated: Sep 28, 2026

Scope: The end-to-end patient journey from hospital entry through intake, records retrieval, consultation, prescription, pharmacy fulfillment, and exit, highlighting operational bottlenecks.



Key Takeaway: Critical operational bottlenecks identified in manual intake queues (25–40 min) and physical prescription transfers.



Operational Gap Analysis

Operational Area	Current State (As-Is)	Desired State (To-Be)	Identified Gap	Action Plan
Patient Registration	Manual paper forms, physical queue tokens.	Digital self-service check-in kiosk & mobile QR check-in.	No digital front-end intake capture.	Deploy touch-screen kiosks with automated token routing.
Medical Records	Physical paper folders transported manually between floors.	Centralized cloud-based EMR with instant lookup.	Decentralized, vulnerable paper records.	Deploy relational database with unified patient ID indexing.
Pharmacy Routing	Handwritten paper prescriptions carried by patients.	Direct electronic push to pharmacy POS terminal.	Manual hand-off latency and handwriting error risks.	Integrate physician order entry with dispensary dashboard.

Target Future-State EMR Workflow

Optimized Hospital Future-State Patient Workflow

Optimized Hospital Future-State Patient Workflow

Owner: Rohit Shorey

Last Updated: Sep 28, 2026

Scope: The optimized future-state patient workflow from arrival through digital check-in, cloud EMR token creation, tablet-based consultation, e-prescription, pharmacy notification, unified checkout, and exit.



Key Takeaway: Automated digital touchpoints eliminate record retrieval latency and reduce prescription fulfillment errors to zero.



System Requirements & MoSCoW Prioritization

Key Requirements:

- **FR-01 (Digital Intake):** Self-service QR check-in & kiosk registration.
- **FR-02 (Central EMR):** Role-based clinician access to consolidated patient history.
- **FR-03 (e-Prescriptions):** Instant digital routing to pharmacy queue.
- **NFR-01 (Security):** End-to-end AES-256 / TLS 1.3 encryption & HIPAA compliance.
- **NFR-02 (Latency):** Sub-1.5 second patient lookup under peak load.

MoSCoW Breakdown:

- **Must Have:** Central EMR database, e-prescribing module, role-based access, data encryption.
- **Should Have:** Automated SMS appointment alerts, real-time inventory deduction at pharmacy.
- **Could Have:** Patient portal for self-service lab report downloads.
- **Won't Have (Phase 2):** Predictive AI diagnosis assistance and automated bed allocation.



Implementation Roadmap & Change Management

Phase 1: Foundation & Training (Weeks 1-2)

- Deploy cloud EMR infrastructure and configure network security.
- Conduct role-based training workshops for doctors, triage nurses, and dispensary staff.

Phase 2: Pilot Dual-Run (Weeks 3-4)

- Launch digital kiosk and e-prescribing in one outpatient department.
- Run parallel paper backups to validate data accuracy and clinician workflow.

Phase 3: Full Cutover & Governance (Week 5+)

- Decommission paper triage slips across all wards.
- Establish weekly operational audits and SLA monitoring.



Target Operational KPIs & Success Metrics

- -40%
- **Patient Wait Time**
- Down from 90+ minutes to <35 minutes total door-to-consultation cycle.

-
- 0%
 - **Prescription Transcription Errors**
 - Complete elimination of legibility-related dispensary delays via e-Prescriptions.

-
- <1.5s
 - **Medical History Retrieval**
 - Instantaneous record lookup replacing 15–20 minute manual file room searches.



Conclusion & Next Steps

Strategic Impact: The proposed architecture resolves acute front-desk bottlenecks and records fragmentation while minimizing administrative overhead for clinical staff.

Feasibility: Non-disruptive phased implementation protects patient continuity of care.

Immediate Next Steps:

1. Sign off on final Functional Specifications Document (FSD).
2. Initiate vendor selection and sprint backlog grooming.
3. Form cross-functional steering committee (Clinical, Operations, IT).